In this page, I investigate some typical finite groups.
Specifically, investigate:
Abelian and Abelization
Commutator and Derived subgroup
Upper-Central Series and Solvablity and Composition Series
Center and Conjugacy classes
Explicit all subgroup

The general properties of S_n, D_n, C_n will be covered in other page.

$\mathbb{Z}_2 \times S_3$, Conjugacy classes.

Since $|G : G_a| = |C_a|$, find the Centralizer of $(0, (12))$.

Since $(1,2)$ commutes only id and itself, and $\mathbb{Z}_2$ is abelian,

$$C_G((0,(1,2))) = \{(0,(12)), (0,id), (1,(1,2)), (1,id)\}.$$

Therefore, the conjugacy classes of $(0,(12))$ has $2 \cdot 6/4 = 3$ elements.

Since $\mathbb{Z}_2$ is abelian (or 0 is 'identity'), the first term never change, and since the conjugate cannot change the cycle structure, the conjugacy classes of $(0,(12))$ is

$$\{(0,(12)), (0,(13)), (0,(23))\},$$

and similarly, the conjugacy classes of $(1,(12))$ is

$$\{(1,(12)), (1,(13)), (1,(23))\},$$

$$\parallel \qquad (0,(123)) \text{ is}$$

$$\{(0,(123)), (0,(132))\}$$

$$\parallel \qquad (1,(123)) \text{ is}$$

$$\{(1,(123)), (1,(132))\}$$

and $(0,id)$ and $(1,id)$ have conjugacy classes only containing itself.

$S_3 \times S_3$, Conjugacy classes.

Let $(1\,2) \times (1\,2) \in S_3 \times S_3$, $(1\,2)$ commute only id or itself, thus

$$C_G((1\,2)) = \{(id, id), (id, (1\,2)), ((1\,2), id), ((1\,2), (1\,2))\},$$

So, the conjugacy classes of $(1\,2) \times (1\,2)$ has 9 elements

S_4.

$S_4 =$	$\{ id,$
1,1,2	$(1\,2), (1\,3), (1\,4), (2\,3), (2\,4), (3\,4),$
2,2	$(1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3),$
1,3	$(1\,2\,3), (1\,3\,2), (1\,3\,4), (1\,4\,3), (1\,2\,4), (1\,4\,2), (2\,3\,4), (2\,4\,3),$
4	$(1\,2\,3\,4), (1\,2\,4\,3), (1\,3\,2\,4), (1\,3\,4\,2), (1\,4\,2\,3), (1\,4\,3\,2)\}$
Partition	

A_4. Conjugacy classes.

$A_4 = \{id, (12)(34), (13)(24), (14)(23),$
$(123), (132), (124), (142), (134), (143),$
$(234), (243)\}$

Since the conjugate cannot change the cycle type,
the conjugacy class of each element in A_4 are contained the set of
same cycle type. But we don't know it exactly equals to the set
let's calculate.

$(123)(12)(34)(123)^{-1} = (23)(14)$

$(124)(12)(34)(124)^{-1} = (24)(31)$

therefore, the conjugacy classes of $(12)(34)$ equals the set of $(2,2)$ cycles.

$(132)(123)(132)^{-1} = (312) = (123)$ (In cuttually, $(123)^{-1} = (132)$)

$(124)(123)(124)^{-1} = (243)$

$(142)(123)(142)^{-1} = (413) = (134)$
$(134)(123)(134)^{-1} = (324) = (243)$
$(143)(123)(143)^{-1} = (421) = (142)$
$(234)(123)(234)^{-1} = (134)$
$(243)(123)(243)^{-1} = (142)$
$(12)(34)(123)(12)(34)^{-1} = (214)$
$(13)(24)(123)(13)(24) = (341) = (134)$.
$(14)(23)(123)(14)(23) = (432) = (243)$

therefore, the conjugacy classes of (123) has just four elements.
Generally, since $|G:G_a| = C_a$ and $G_a = C_G(a)$ in the conjugate action,
therefore number of elements of the orbit is $|G|/|C_G(a)|$.
Since $C_{A_4}((123)) = \{id, (123), (132)\}$,

the orbit of (123) has $12/3 = 4$ elements.

A_5. The smallest group of non-abelian simple group.

Prop. Group of order 60 having more than one Sylow 5-subgroup is simple.

Proof. Using Contradiction. Suppose that there is a normal subgroup $1 < H \triangleleft G$.
By the Sylow Theorem, $n_5 \equiv 1 \bmod 5$ and $n_5 \mid 12$, so $n_5 \in \{1, 6\}$.
By the hypothesis, n_5 must be 6. Let $P \in Syl_5(G)$ be a Sylow 5-subgroup.
Since $n_5 = 6$, that is, $|\{gPg^{-1} \mid g \in G\}| = |G : N_G(P)| = 6$, therefore $|N_G(P)| = 10$.
Assume that $5 \mid |H|$. Then, H contains a Sylow 5-subgroup Q of G,
because $60 = 2^2 \cdot 3 \cdot 5$, and since H is Normal $gQg^{-1} \subseteq H$ for all $g \in G$,
therefore H contains all Sylow 5-subgroups of G.
But, Since for any $P_1, P_2 \in Syl_5(G)$, $P_1 \neq P_2 \Rightarrow P_1 \cap P_2 = \{e\}$,
therefore $|H| \geq 5 \cdot 6 - 6 \cdot 1 + 1 = 25$, so the Lagrange's Theorem forces $|H| = 30$.
But, by the Sylow Theorem, $\#Syl_3(H) \in \{1, 10\}$. If $\#Syl_3(H) = 1$,
then $A \in Syl_3(H)$ must be a normal subgroup of H, then PA is a group of
order $|PA| = |P| \cdot |A| / |P \cap A| = 15/1 = 15$, so $|H : PA| = 30/15 = 2$, $PA \triangleleft H$ is normal.
But, order of 5 subgroup in group of 15 is characteristic, because $n_5 \mid 3 \Rightarrow n_5 = 1$,
so P characteristic to PA, and $PA \triangleleft H$, so $P \triangleleft H$. It contradicts.
Thus, $\#Syl_3(H) = 10$, but every non-identity element of Sylow 3-subgroup has
order 3, and every non-identity element of Sylow 5-subgroup has order 5, so

$$|H| \geq 6 \cdot 4 + 10 \cdot 2 = 24 + 20 = 44$$

But it cannot be happen since $|H| = 30$. So, $|H|$ is not divided by 5.
Now, $|H| \in \{2, 4, 6, 12\}$. If $|H| = 6$, then by the sylow theorem,
there is $H_1 < H$ such that $H_1 \in Syl_3(H)$. If $|H| = 12$, then for $n_3 = \#Syl_3(H)$,
$n_3 = 1$ or $n_3 = 4$. But, $n_3 = 4 \Rightarrow |H| \geq n_3 \cdot (3-1) + 1 + n_2 \cdot (4-1) = 9 + 3 \cdot n_2$
$\Rightarrow n_2 = 1$
So there is $H_2 < H$ such that H_2 char H. Anyway, H_1, H_2 char H,
so $H_1, H_2 \triangleleft G$, Now, $|G/H_1|, |G/H_2| \in \{15, 10\}$, So $|G/H_i|$ $i = 1, 2$. has
a normal subgroup $K/H_i \triangleleft G/H_i$. Now, by the corresponding isomorphism theorem,
$K \triangleleft G$ and $5 \mid |K|$, it contradicts by the above argument.

Klein 4-group.

Let $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ be the Klein 4-group.

Then, a map $\varphi: G \to S_4$:

$(0,0) \mapsto id$, $(1,0) \mapsto (12)(34)$, $(0,1) \mapsto (13)(24)$, $(1,1) \mapsto (14)(23)$

is Embedding

Let p be an odd prime number. Let $G = D_{2p} \times C_p$, where D is Dihedral, C is cyclic.

1). Find all sylow subgroup of G.

2). Find all p-subgroup.

3). Find the commutator subgroup of G and abelization.

4). Find $Aut(G)$.

Proof. Since $|G| = |D_{2p}| \cdot |C_p| = 2p^2$, by the sylow theorem, $n_p = \#Syl_p(G)$ satisfies that $n_p \equiv 1 \bmod p$ and $n_p \mid 2$, so that $n_p = 1$. Denote $C_p = \langle x \rangle$. Denote that $D_{2p} = \langle r, s \mid r^p = s^2 = (rs)^2 = e \rangle$. Then, $\langle r \rangle < D_{2p}$ with $|\langle r \rangle| = p$, therefore the subgroup $\langle r \rangle \times C_p < G$ is order of p^2, so it is the unique sylow p-subgroup. Meanwhile, by the sylow theorem again, $n_2 \equiv 1 \bmod 2$ and $n_2 \mid p^2$, so $n_2 \in \{1, p, p^2\}$. But, since $\langle s \rangle \times \{x^0\}$ and $\langle rs \rangle \times \{x^0\}$ are order 2 subgroups of G, so $n_2 > 1$. If $n_2 = p^2$, then there are p^2-numbers distict order 2 elements, but $(a, x^k) \in G$ has order 2 iff $k = 0$ and $|a| = 2$, but $\{r, r^2, \cdots, r^{p-1}\} \subset D_{2p}$ have order of p, so D_{2p} cannot have p^2-numbers order 2 elements, by just counting. (In fact, $2p < p^2$ also gives same result). So, G has p-numbers sylow 2-subgroups,

$$\langle rs \rangle \times \{x^0\},\ \langle s \rangle \times \{x^0\},\ \langle r^2 s \rangle \times \{x^0\},\ \cdots,\ \langle r^{p-1} s \rangle \times \{x^0\}.$$

2). By the sylow theorem, all p-subgroups are contained for some sylow p-subgroup, but since G has a unique sylow p-subgroup, $\langle r \rangle \times C_p$, so all p-subgroups are $\langle r \rangle \times C_p$ itself and

$\langle r \rangle \times \{x^0\}$, $\{e\} \times C_p$, $\langle (r^k, x^l) \rangle$ $k = 1, 2, \cdots, p-1$,

($\langle r \rangle \times C_p \cong C_p \times C_p$, it has $p^2 - 1$ elements of order of p, $\cdots$)

3). Since the operation on $G = D_{2p} \times C_p$ was defined by entrywise, so that $[G, G] = [D_{2p}, D_{2p}] \times [C_p, C_p]$. Since C_p is abelian, $[C_p, C_p] = \{x^0\}$, and $[D_{2p}, D_{2p}] = \langle r \rangle$, so that the commutator is $\langle r \rangle \times \{x^0\}$, Consequently, the abelization is $D_{2p} \times C_p / \langle r \rangle \times \{x^0\} \cong C_2 \times C_p$.

4).